

Phromsawat Phoolprom

B.Sc. Geography & Geoinformatics | Srinakharinwirot University

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EDUCATION

BACHELOR OF SCIENCE IN GEOGRAPHY & GEOINFORMATICS

Bangkok, TH

Srinakharinwirot University - GPA 3.09

May 2026

CORE TECHNICAL COMPETENCIES

- Programming & Web Frameworks:** TypeScript, Next.js, Python, FastAPI, Tailwind CSS
- Geospatial & Mapping APIs:** ArcGIS Pro, QGIS, PostGIS, Geopandas, Leaflet, Google Maps API, Google Earth Engine, ArcGIS Field Maps
- Data Science & Analytics:** PostgreSQL, Supabase, Pandas, NumPy, Matplotlib, Spatial Statistics, Network Analysis
- Developer Tools & Workflows:** Docker, Git, Figma, OSM Editing Tools, AI-Assisted Development (LLM Code Agents)

PROFESSIONAL EXPERIENCE

DEPARTMENT OF AGRICULTURE (DOA)

Bangkok, TH

Full Stack Web Developer — Soil Analysis & Fertilizer Recommendation System

APR 2026 – Present

- Engineered** a responsive Next.js frontend for soil nutrient analysis, converting deep learning predictions into real-time interactive dashboards and charts.
- Designed** and integrated secure REST APIs using FastAPI and PostgreSQL to fetch and render soil profiles.

SRINAKHARINWIROT UNIVERSITY

Bangkok, TH

Freelance GIS Data Specialist

MAR 2025

- Constructed** a campus building footprint database in QGIS for facility management.

DEPARTMENT OF WATER RESOURCES

Bangkok, TH

Geospatial Data Analyst (Project-based)

APR – DEC 2025

- Compiled** multi-source spatial datasets and generated flood simulation maps for the Mun-Chi river basin.
- Programmed** geoprocessing and attribute validation workflows using Python (ArcPy) and SQL in ArcGIS Pro, using AI-assisted development to accelerate pipeline delivery.

I-BITZ COMPANY LIMITED

Bangkok, TH

UX/UI Design & Front-End Intern — MoodMap Project

AUG – NOV 2025

- Developed** interactive mapping components using Next.js, Tailwind CSS, and Leaflet/Google Maps APIs for high-performance spatial rendering.
- Created** a unified design system and interactive wireframes with 100+ screens in Figma based on user flow research.

SRINAKHARINWIROT UNIVERSITY

Bangkok & Nakhon Nayok, TH

GIS Field Researcher & Trainer (Project-based)

MAY 2024 – JUL 2025

- Analyzed** biomass and carbon sequestration using field data collected via ArcGIS Field Maps.
- Trained** university staff on mobile GIS data collection workflows.

ELECTRICITY GENERATING AUTHORITY OF THAILAND &

Bangkok, TH

SIIT, THAMMASAT UNIVERSITY

GIS Data Specialist (Project-based)

JUN – OCT 2024

- Classified** 2,000+ solar panel boundaries from satellite imagery for U-Net deep learning model training, validating detection accuracy via ground-truth surveys.

SELECTED PROJECT EXPERIENCE

- **Live Demo:**paipai-trip-planner.vercel.app
- **Built** a responsive map-centric frontend in **Next.js 16 (React 19)**, integrating **Leaflet** with a glass-morphism UI for itinerary visualization.
- **Implemented** client-side walking speed configurations, real-time AQI (PM2.5), and weather alert overlays (Open-Meteo) on interactive map layers.
- **Devised** a persistent two-tier caching system (segment RAM/DB & JSONB plan cache) with PostGIS indexing, reducing API latency from 10s to <100ms.
- **Developed** a FastAPI routing service using **Google OR-Tools** (Guided Local Search) for TSP route optimization.
- **Expedited** full-stack development by orchestrating **AI coding agents** (Antigravity/Claude), conducting human-in-the-loop code reviews to ensure system security and clean architecture.

- **Evaluated** and compared **Nostra Map** and **Longdo Map** API routing accuracy through field testing in high-density urban corridors.

- **Modeled** industrial chemical dispersion in **ALOHA** and conducted network analysis to map safe zones and evacuation corridors.

LEADERSHIP & VOLUNTEER ENGAGEMENT

- **[Jun 2025]** Disaster Response Mapper: **Compiled** Hat Yai building and road networks for emergency flood relief.
- **[Jun 2025]** YouthMappers Asia Contributor: **Documented** Hua Hin building footprints during regional networking events.
- **[2021 – Present]** OSM Editor: **Contributed** to OpenStreetMap by compiling Thai road networks and footprints.

- **[Dec 2024 & Nov 2023]** Staff Coordinator: **Coordinated** event logistics and facilitated communication for international attendees in English.